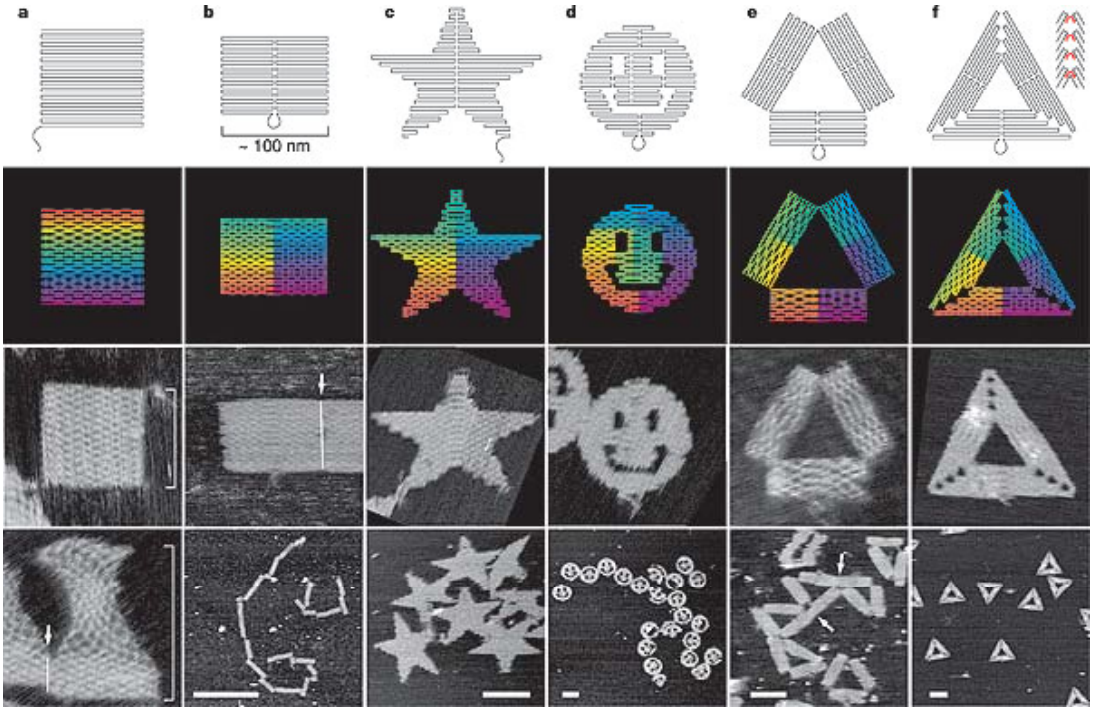
화학특강 (2012년 2학기)
제2장 DNA, 단백질 그리고 Biotechnology

중요 학습 내용

- DNA의 구조
- 단백질의 구조와 합성
- DNA translocation and DNA transfection
- Colorimetric sensing

DNA origami



Rothemund, Nature **440**, 297 (2006)

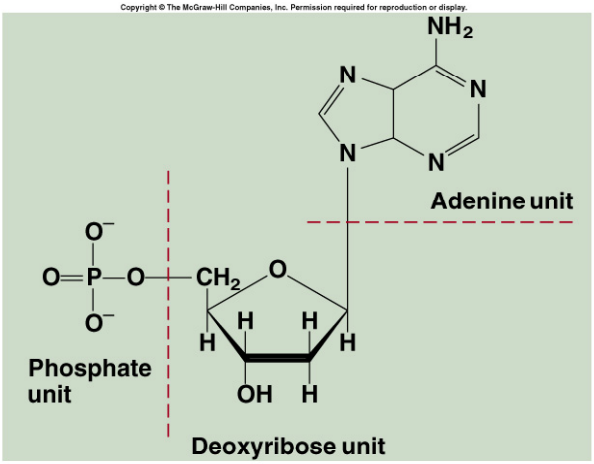
Self assembly of ssDNA



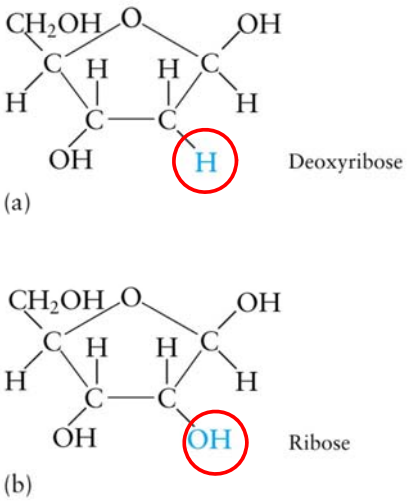
Wei et al., Nature **485**, 623 (2012)

Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

- Structure of Nucleotide



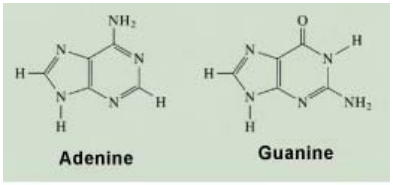
- Deoxyribose and ribose



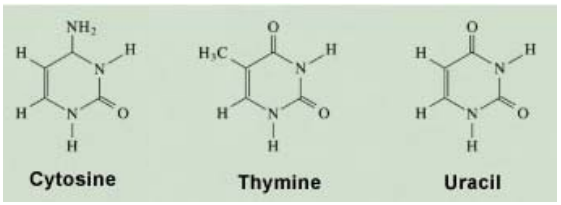
Chang, 25.4

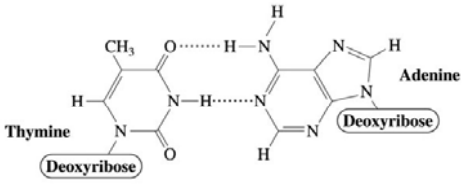
Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

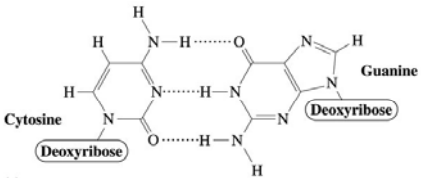
- Purine bases



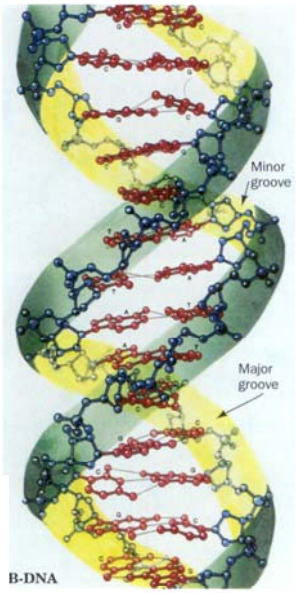
- Pyrimidine bases







- DNA double helix

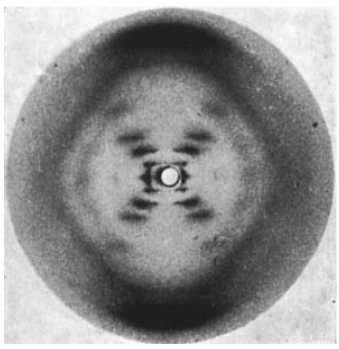


Chang, 25.4

History of DNA

- Chargaff's rules in the 1940's
  1. The amount of adenine (a purine) is equal to that of thymine (a pyrimidine); that is A = T, or A/T = 1.
  2. The amount of guanine (a purine) is equal to that of cytosine (a pyrimidine); that is G = C, or G/C = 1.
  3. The total number of purine bases is equal to the total number of pyrimidine bases; that is, A + G = T + C .

- X-ray diffraction photograph by R. Franklin





Chang, 25.4

History of DNA

- Nature paper by Watson & Crick



MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons:

(1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β-D-deoxyribofuranose residues with 3′,5′ linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow right-handed helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions.

Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain, does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain, is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in time following, communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J.D.W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

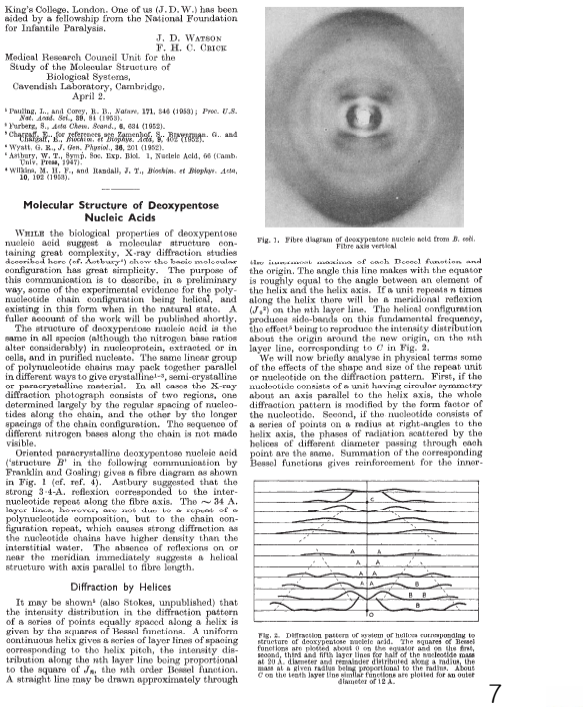
J.D. WATSON
F.H.C. CRICK

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

[1] Pauling, L., and Corey, R. B., *Nature*, **171**, 346 (1953); *Proc. U.S. Nat. Acad. Sci.*, **39**, 84 (1953).
[2] Furberg, S., *Acta Chem. Scand.*, **6**, 634 (1952).
[3] Chargaff, E., *Experientia*, **6**, 201 (1950).
[4] Wyatt, G. R., *J. Gen. Physiol.*, **36**, 201 (1952).
[5] Astbury, W. T., *Symp. Soc. Exp. Biol. 1, Nucleic Acid*, 66 (Camb. Univ. Press, 1947).
[6] Wilkins, M. H. F., and Randall, J. T., *Biochim. et Biophys. Acta*, **10**, 192 (1953).

Watson & Crick, Nature 171, 737 (1953)

- Nature paper by Wilkins



Molecular Structure of Deoxypentose Nucleic Acids

WHILE the biological properties of deoxypentose nucleic acid suggest a molecular structure containing great complexity, X-ray diffraction studies described here (cf. Astbury[1]) show the basic molecular configuration has great simplicity. The purpose of this communication is to describe, in a preliminary way, some of the experimental evidence for the polynucleotide chain configuration being helical, and existing in this form when in the natural state. A fuller account of the work will be published shortly.

The structure of deoxypentose nucleic acid is the same in all species (although the nitrogen base ratios alter considerably) in nucleoprotein, extracted or in cells, and in purified nucleate. The same linear group of polynucleotide chains may pack together parallel in different ways to give crystalline[1-3], semi-crystalline or paracrystalline material. In all cases the X-ray diffraction photograph consists of two regions, one determined largely by the regular spacing of nucleotides along the chain, and the other by the longer spacings of the chain configuration. The sequence of different nitrogen bases along the chain is not made visible.

Oriented paracrystalline deoxypentose nucleic acid ('structure *B*' in the following communication by Franklin and Gosling) gives a fibre diagram as shown in Fig. 1 (cf. ref. 4). Astbury suggested that the strong 3·4-A. reflexion corresponded to the internucleotide repeat along the fibre axis. The ~ 34 A. layer lines, however, are not due to a repeat of a polynucleotide composition, but to the chain configuration repeat, which causes strong diffraction as the nucleotide chains have higher density than the interstitial water. The absence of reflexions on or near the meridian immediately suggests a helical structure with axis parallel to fibre length.

Diffraction by Helices

It may be shown[4] (also Stokes, unpublished) that the intensity distribution in the diffraction pattern of a series of points equally spaced along a helix is given by the squares of Bessel functions. A uniform continuous helix gives a series of layer lines of spacing corresponding to the helix pitch, the intensity distribution along the nth layer line being proportional to the square of J_n, the nth order Bessel function. A straight line may be drawn approximately through the innermost maxima of the first, second, third and fifth layer lines, which are approximately on straight lines radiating from the origin. For a single helix, the intensity maxima of the type observed, corresponding to the first maxima in the functions J_1, J_2, J_3, etc. If, instead of a smooth helix, we consider a series of residues equally spaced along the helix, the transform in the general case treated by Crick, Cochran and Vand is more complicated. But if there is a whole number, *m*, of residues per turn, the form of the transform is as for a smooth helix with the addition, only, of the same pattern repeated with its origin at heights $m\sigma$, $2m\sigma$, . . . etc. (σ is the fibre-axis period).

In the present case the fibre-axis period is 34 A. and the very strong reflexion at 3·4 A. lies on the tenth layer line. Moreover, lines of maxima radiating from the 3·4-A. reflexion as from the origin are visible on the fifth and lower layer lines, having a J_4 maximum coincident with that of the origin series on the fifth layer line. (The strong outer streaks which apparently radiate from the 3·4-A. maximum are not, however, so easily explained.) This suggests strongly that there are exactly 10 residues per turn of the helix. If this is so, then from a measurement of R_0 the position of the first maximum on the nth layer line (for n ≠ 0), the radius of the helix can be obtained. In the present instance, measurements of R_1, R_2, R_3 and R_4 all lead to values of *r* of about 10 A.

Fig. 1. Fibre diagram of deoxypentose nucleic acid from *B. coli*. Fibre axis vertical

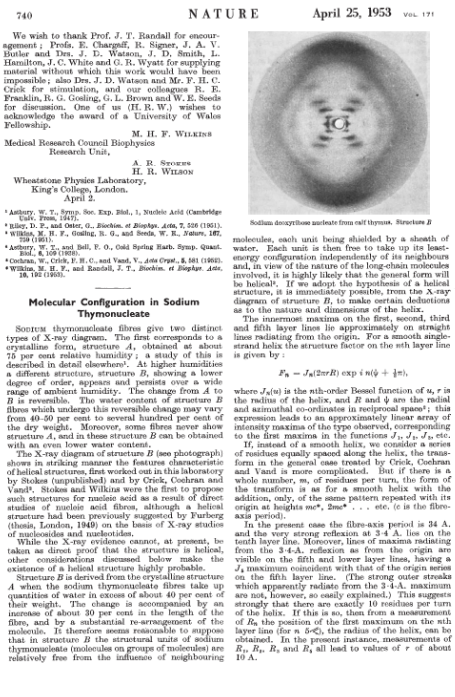
Fig. 2. Diffraction pattern of system of helices corresponding to structure of deoxypentose nucleic acid. The squares of Bessel functions are plotted about *O* on the equator and on the first, second, third and fifth layer lines for half of the nucleotide mass at 10 A. diameter and remainder distributed along a radius, the mass at a given radius being proportional to the radius. About *O* on the tenth layer line similar functions are plotted for an outer diameter of 12 A.

©1953 Nature Publishing Group

Wilkins, Stoke, and Wilson, Nature 171, 738 (1953)

History of DNA

- Nature paper by Rosalind Franklin



Molecular Configuration in Sodium Thymonucleate

SODIUM thymonucleate fibres give two distinct types of X-ray diagram. The first corresponds to a crystalline form, structure *A*, obtained at about 75 per cent relative humidity; a study of this is described in detail elsewhere[1]. At higher humidities a different structure, structure *B*, showing a lower degree of order, appears and persists over a wide range of ambient humidity. The change from *A* to *B* is reversible. The water content of structure *B* fibres which undergo this reversible change may vary from 40–50 per cent to several hundred per cent of the dry weight. Moreover, some fibres never show structure *A*, and in these structure *B* can be obtained with an even lower water content.

The X-ray diagram of structure *B* (see photograph) shows in striking manner the features characteristic of helical structures, first worked out in this laboratory by Stokes (unpublished) and by Crick, Cochran and Vand[2]. Stokes and Wilkins were the first to propose such structures for nucleic acid as a result of direct studies of nucleic acid fibres, although a helical structure had been previously suggested by Furberg (thesis, London, 1949) on the basis of X-ray studies of nucleosides and nucleotides.

While the X-ray evidence cannot, at present, be taken as direct proof that the structure is helical, other considerations discussed below make the existence of a helical structure highly probable.

Structure *B* is derived from the crystalline structure *A* when the sodium thymonucleate fibres take up quantities of water in excess of about 40 per cent of their weight. The change is accompanied by an increase of about 30 per cent in the length of the fibre, and by a substantial re-arrangement of the molecule. It therefore seems reasonable to suppose that in structure *B* the structural units of sodium thymonucleate (molecules or groups of molecules) are relatively free from the influence of neighbouring molecules, each unit being shielded by a sheath of water. Each unit is then free to take up its least-energy configuration independently of its neighbours and, in view of the nature of the long-chain molecules involved, it is highly likely that the general form will be helical[3]. If we adopt the hypothesis of a helical structure, it is immediately possible, from the X-ray diagram of structure *B*, to make certain deductions as to the nature and dimensions of the helix.

Sodium deoxyribose nucleate from calf thymus. Structure *B*

The innermost maxima on the first, second, third and fifth layer lines lie approximately on straight lines radiating from the origin. For a single-strand helix the structure factor on the *n*th layer line is given by:

$$F_n = J_n(2\pi rR)\exp i n(\psi + \tfrac{1}{2}\pi),$$

where $J_n(u)$ is the *n*th-order Bessel function of u, r is the radius of the helix, and R and ψ are the radial and azimuthal co-ordinates in reciprocal space[4]; this expression leads to an approximately linear array of intensity maxima of the type observed, corresponding to the first maxima in the functions J_1, J_2, J_3, etc.

In the present case the fibre-axis period is 34 A. and the very strong reflexion at 3·4 A. lies on the tenth layer line. Moreover, lines of maxima radiating from the 3·4-A. reflexion as from the origin are visible on the fifth and lower layer lines, having a J_4 maximum coincident with that of the origin series on the fifth layer line. (The strong outer streaks which apparently radiate from the 3·4-A. maximum are not, however, so easily explained.) This suggests strongly that there are exactly 10 residues per turn of the helix. If this is so, then from a measurement of R_0 the position of the first maximum on the *n*th layer line (for $n \neq 0$), the radius of the helix can be obtained. In the present instance, measurements of R_1, R_2, R_3 and R_4 all lead to values of *r* of about 10 A.

©1953 Nature Publishing Group

Franklin & Gobling, Nature 171, 740 (1953)



The Nobel Prize in Physiology or Medicine 1962

"for their discoveries concerning the molecular structure of nucleic acids and its significance for information transfer in living material"



Francis Harry Compton Crick

1/3 of the prize

United Kingdom

MRC Laboratory of Molecular Biology
Cambridge, United Kingdom



James Dewey Watson

1/3 of the prize

USA

Harvard University
Cambridge, MA, USA

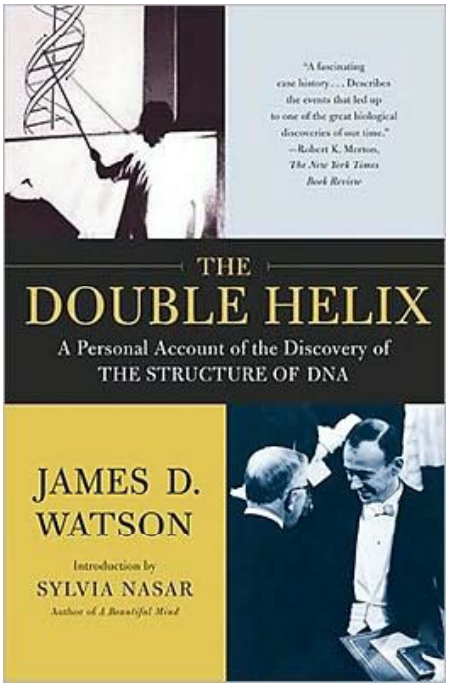


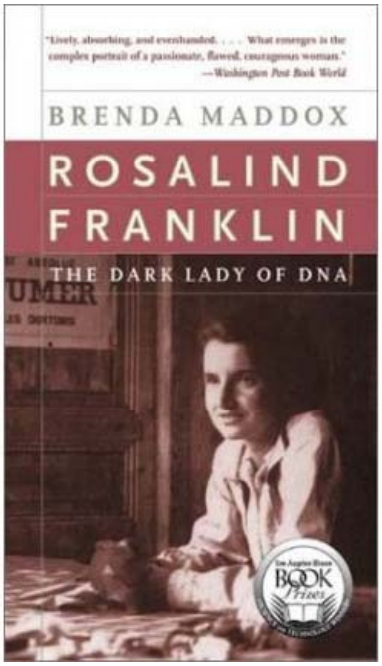
Maurice Hugh Frederick Wilkins

1/3 of the prize

United Kingdom and New Zealand

London University
London, United Kingdom





Amino Acids

- **Amino acids**: the basic structural units of proteins. An amino acid is a compound that contains at least one amino group (-NH2) and at least one carboxyl group (-COOH)

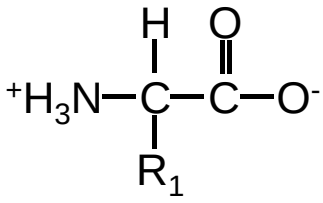
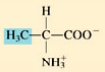
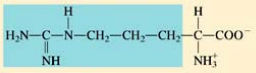
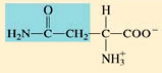
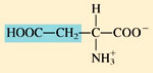
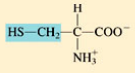
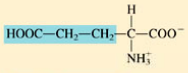
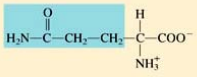
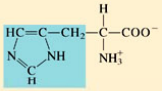
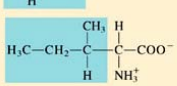


TABLE 25.1 The 20 Amino Acids Essential to Living Organisms*

Name	Abbreviation	Structure
Alanine	Ala	
Arginine	Arg	
Asparagine	Asn	
Aspartic acid	Asp	
Cysteine	Cys	
Glutamic acid	Glu	
Glutamine	Gln	
Glycine	Gly	
Histidine	His	
Isoleucine	Ile	

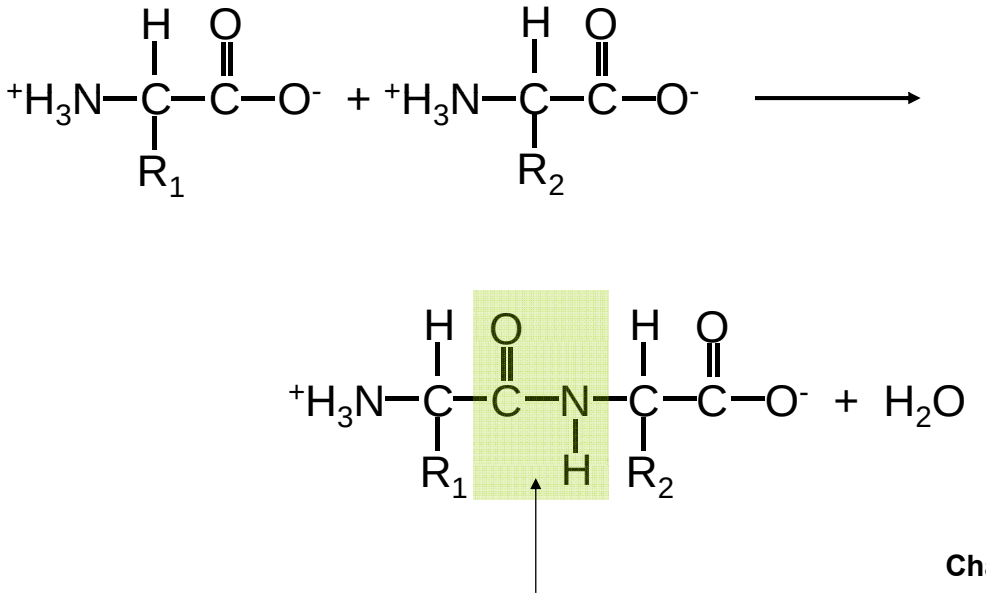
(Continued)

*The shaded portion is the R group of the amino acid.

Chang, 25.4

Proteins

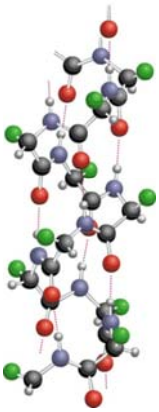
- **Proteins**: A polymer of amino acids through peptide bonds.



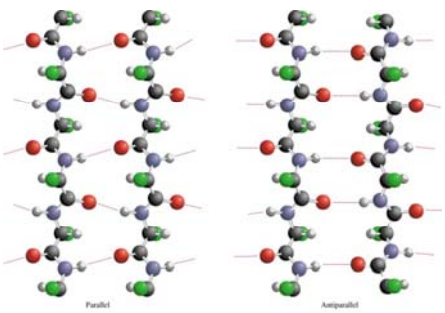
Chang, 25.4

Protein Structure

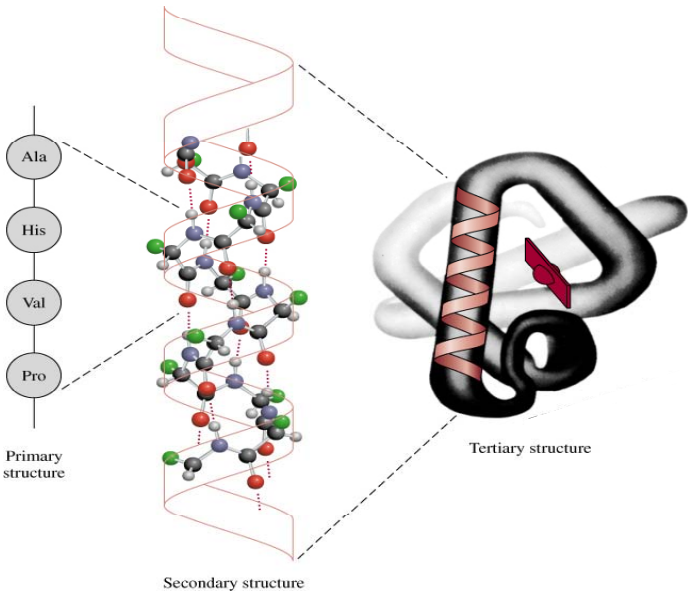
- α helix



- β sheet



- Primary, secondary, and tertiary structures

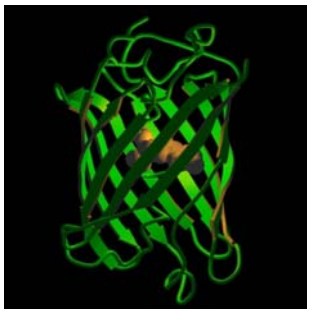


Chang, 25.4

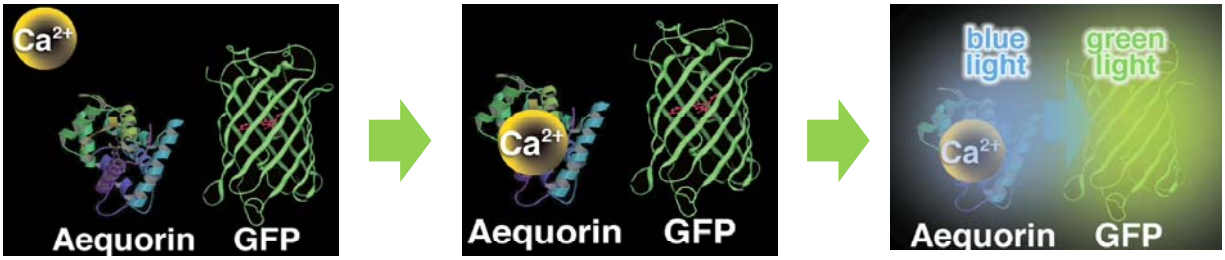
What is Green Fluorescent Protein (GFP)?



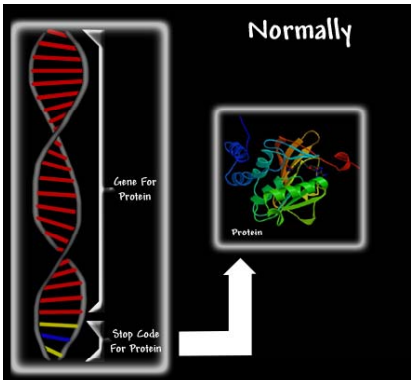
Aequorea victoria jellyfish

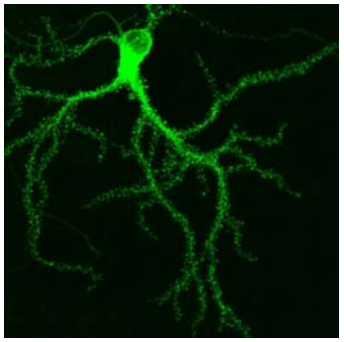


GFP

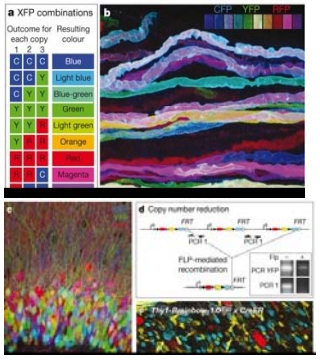


Applications of GFP

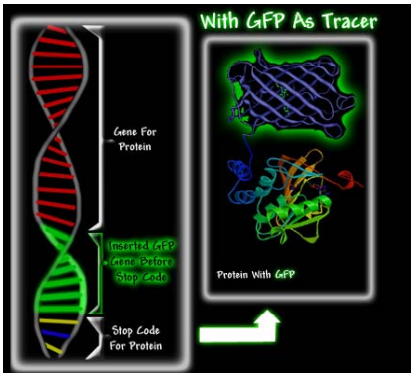




Neuron GFP

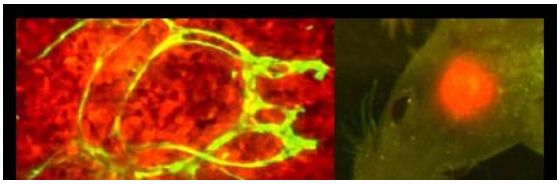


Livet et al., Nature **450**, 56 (2007)





GFP cloned 쥐와 보통 쥐



암세포들 (붉은색)과 혈관 (녹색)

http://www.conncoll.edu/ccacad/zimmer/GFP-ww/GFP-1.htm

The Nobel Prize in Chemistry 2008

"for the discovery and development of the green fluorescent protein, GFP"

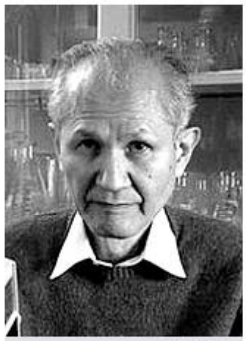


Photo: J. Henriksson/SCANPIX

Osamu Shimomura

1/3 of the prize

USA

Marine Biological Laboratory (MBL)
Woods Hole, MA, USA

b. 1928



Photo: J. Henriksson/SCANPIX

Martin Chalfie

1/3 of the prize

USA

Columbia University
New York, NY, USA

b. 1947



Photo: UCSD

Roger Y. Tsien

1/3 of the prize

USA

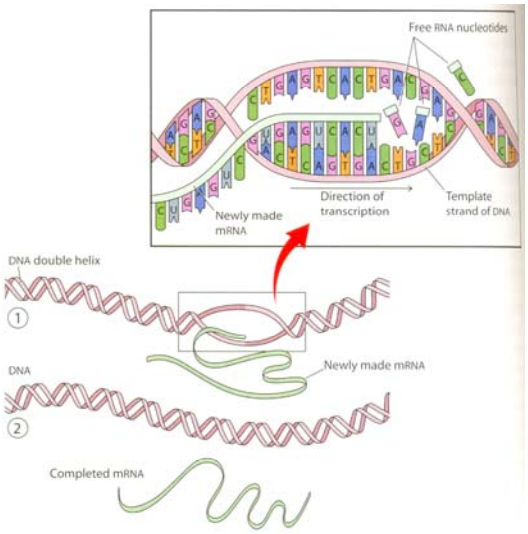
University of California
San Diego, CA, USA

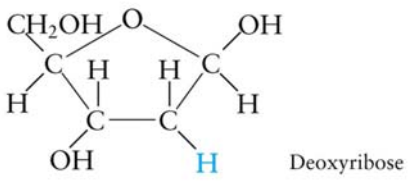
b. 1952

KNU

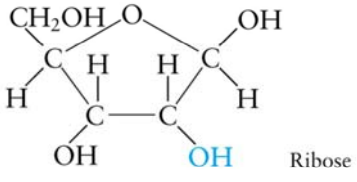
Protein Synthesis: Transcription

- DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ protein
- Transcription takes place in the cell nucleus





(a)

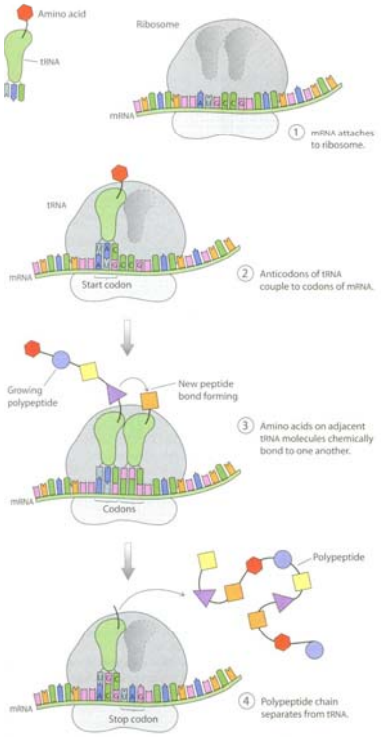


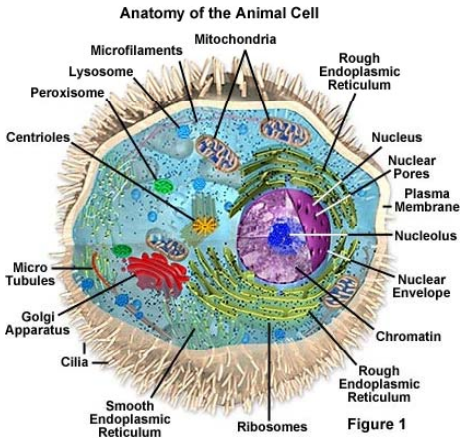
(b)

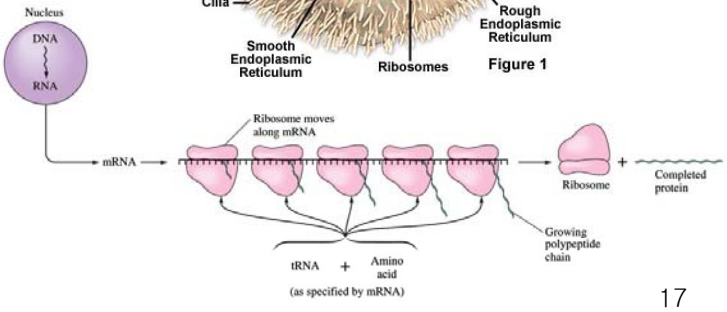
KNU

Protein Synthesis: Translation

- Ribosomes are microscopic complexes of two-thirds rRNA and one-third proteins, and they are the site where proteins are built.





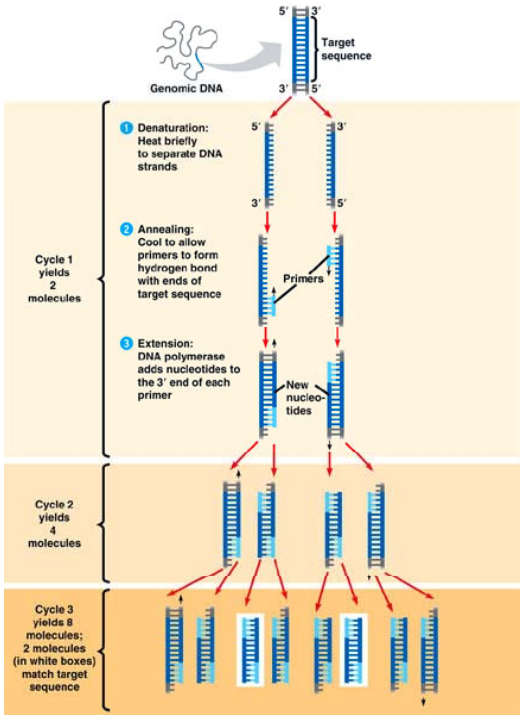


Polymerase Chain Reaction (PCR)

- **PCR**
  - A technique to amplify a piece of DNA
  - DNA template (target), two primers, DNA polymerase (Taq polymerase), deoxinucleoside triphosphates.

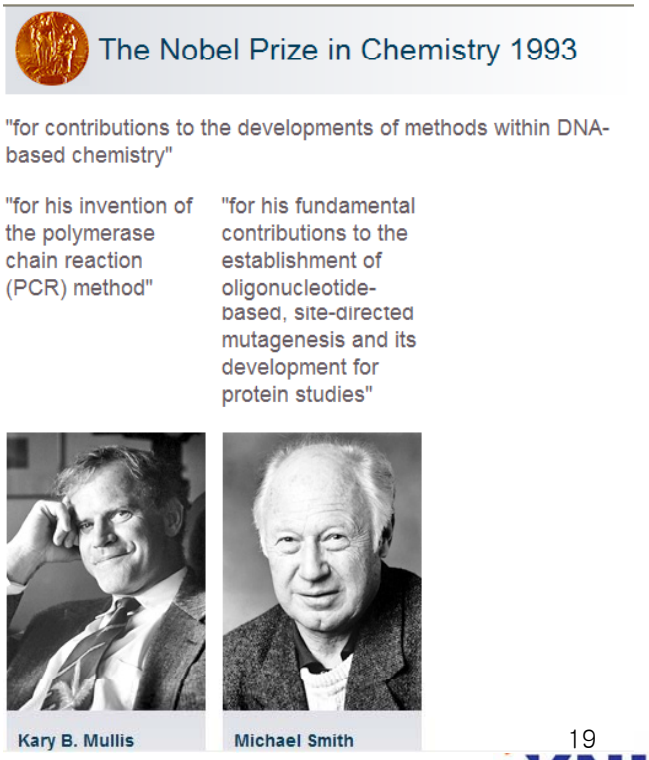
- Procedure
  - Initialization: heating to 94-96°C
  - Denaturation: yielding ssDNAs
  - Annealing: lowering to 50-65°C and annealing primers to ssDNAs
  - Elongation: inducing DNA polymerases to synthesize a new DNA strand

- Applications of PCR
  - Isolation of genomic DNA
  - DNA sequencing
  - Forensic analysis
  - Disease diagnosis

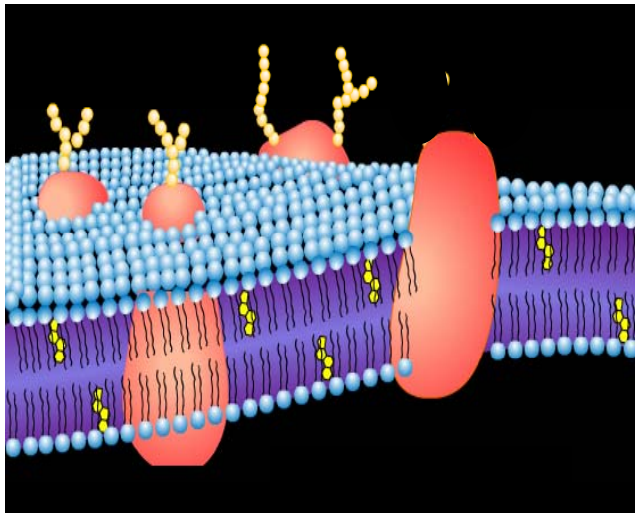


Polymerase Chain Reaction (PCR)

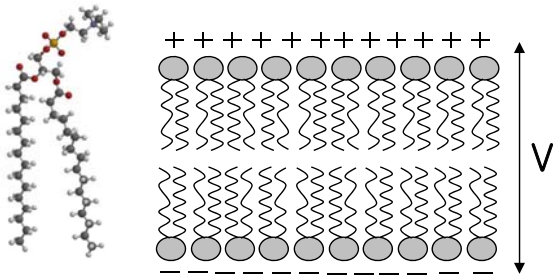


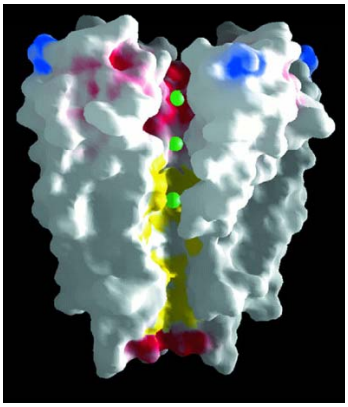


What are ion channels?



Voet and Voet, Biochemistry (3rd. Ed.)





KcsA ion channel
Science 280, 69 (1998)

What are ion channels?



Black mamba



Dendrotoxin

Chemistry

The Nobel Prize in Chemistry 2003

"for discoveries concerning channels in cell membranes"

"for the discovery of water channels"

"for structural and mechanistic studies of ion channels"



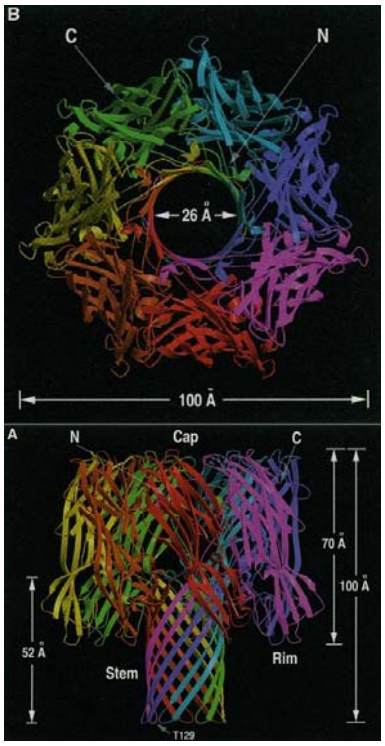
Peter Agre



Roderick MacKinnon

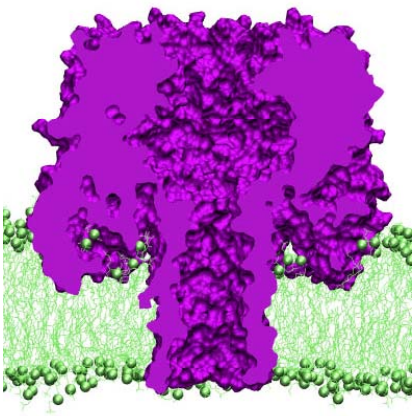
DNA translocation and alpha hemolysin

- Alpha hemolysin: an ion channel

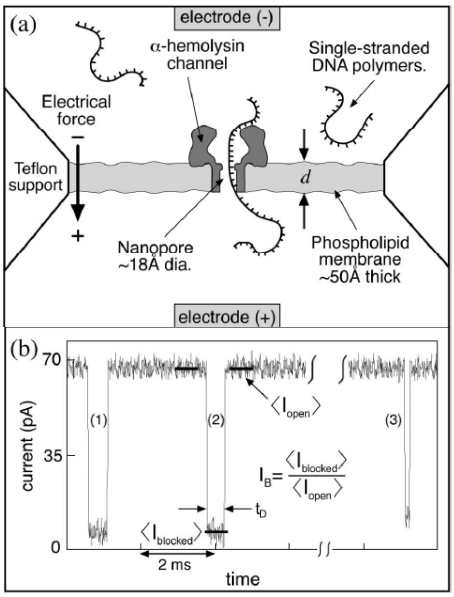


52 Å

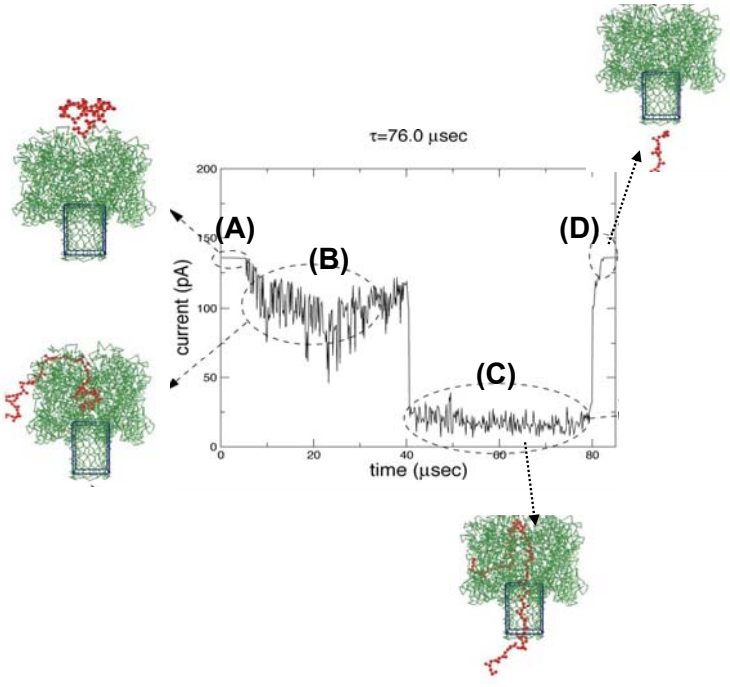
- Alpha hemolysin inside a lipid bilayer



ssDNA translocations in experiments and simulations



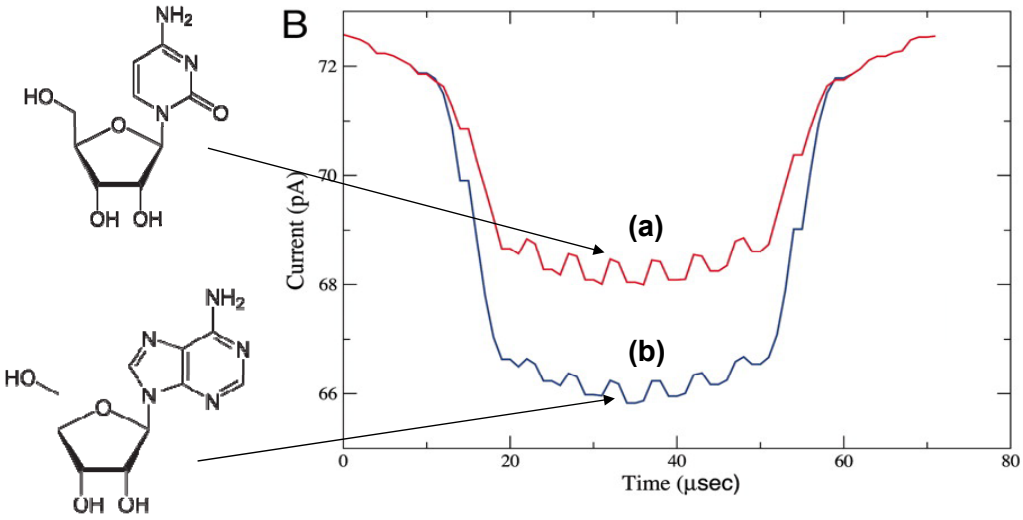
Miller et al., PRL **86**, 3435 (2001)



Muthukumar et al., PNAS **103**, 5273 (2006)

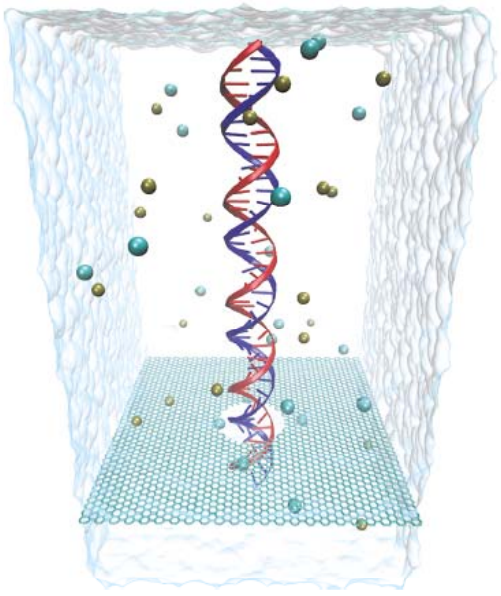


(예제) ssDNA를 구성하는 염기의 종류에 따라 ssDNA가 이온채널을 통과할 때 나타나는 이온전류의 변화는 달라진다. 컴퓨터 모의실험에서 adenine 염기로만 이루어진 ssDNA (poly-dA)와 cytosine 염기로만 이루어진 ssDNA (poly-dC)를 이용하여 각각 모의실험을 하였더니 다음과 같은 시간-이온전류 변화 그래프를 얻었다고 할 때, (a)와 (b)에 해당하는 ssDNA는 무엇인가? 그 이유를 설명하여라.



Muthukumar et al., PNAS **103**, 5273 (2006)

dsDNA translocation through a graphene by molecular dynamics (MD) simulations



jure 1. Atomic model of the graphene nanopore system nulated in this study. Shown is dsDNA in its initial upright sition inside a graphene nanopore of 2.4 nm diameter; o shown are K^+ and Cl^- ions, as well as the water surface the boundaries of the simulated periodic cell (96 Å × 96 Å 220 Å).

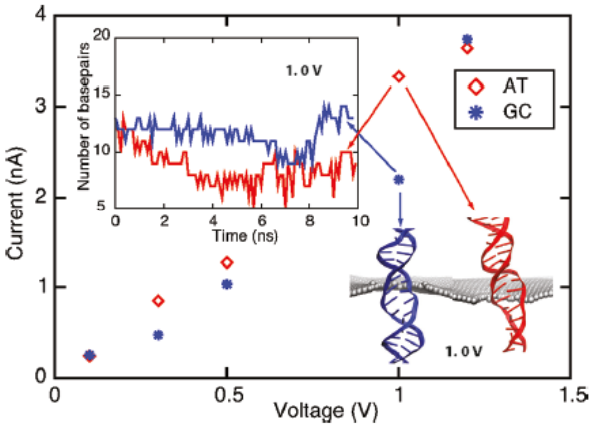
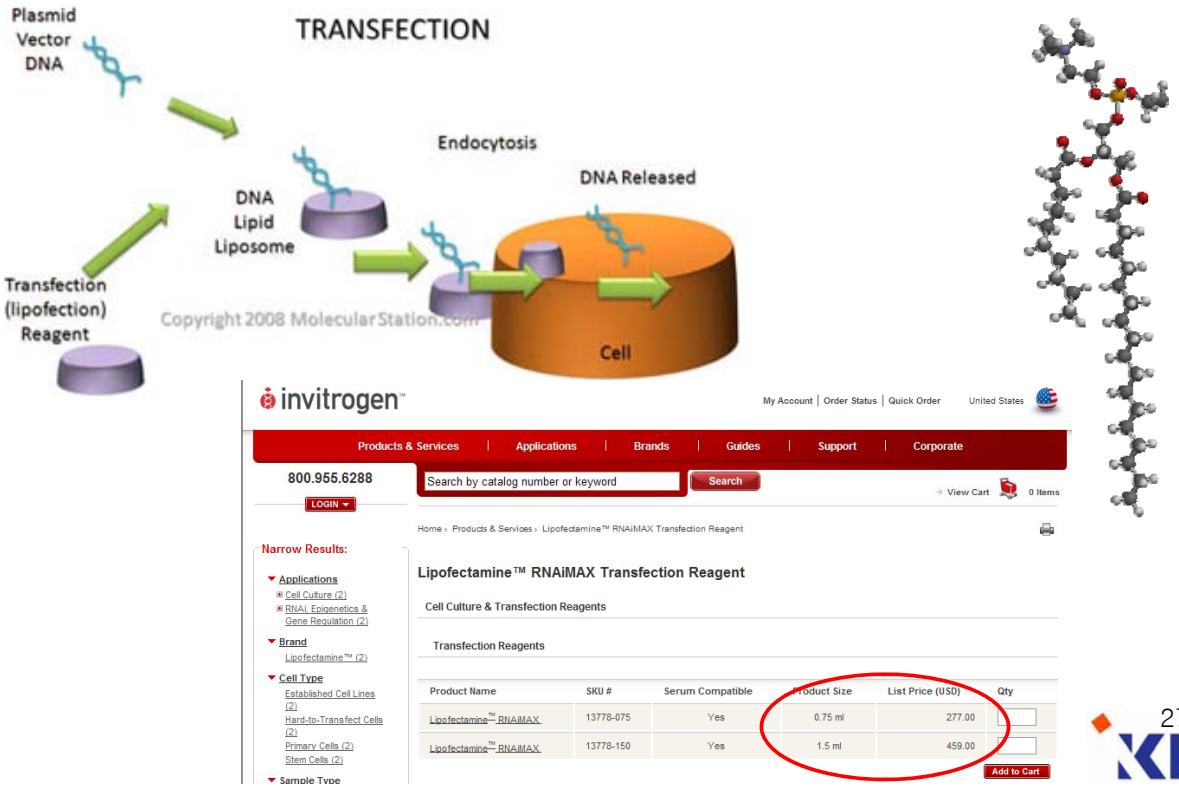


Figure 6. Ionic current for poly(A-T)$_{20}$ and poly(G-C)$_{20}$ duplexes measured at 0.1, 0.3, 0.5, 1.0, and 1.2 V transmembrane bias voltages in a 2.4 nm diameter nanopore (SimE1−SimF5). Translocation of A-T and G-C base pairs results in different ionic currents at 0.3, 0.5, and 1.0 V. Snapshot shows that poly(A-T)$_{20}$ (red) is more stretched and disordered than poly(G-C)$_{20}$ (blue) at 1.0 V (see Supporting Information movies S6 and S7). The inset shows the number of base pairs near the pore mouth (±2 nm): A-T base pairs are more readily broken than G-C base pairs at 1.0 V. Figure S4 shows the number of base pairs near the pore mouth for 0.1, 0.3, 0.5, and 1.2 V.

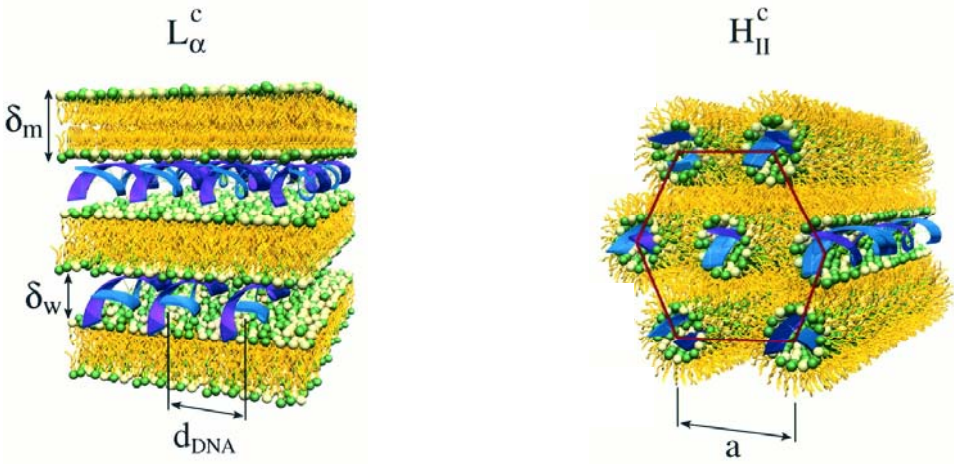
Sathe, C. et al. *ACS Nano*. **5**, 8842 (2011)

DNA Transfection



Lipid phases in lipid-DNA complex (=lipoplex)

- Lamellar(L_α^c) and hexagonal(H_{II}^c) phases.



(Science, **281**, 78 (1998))

Beer-Lambert Law

- Derivation of Beer-Lambert Law

$$dI = -\kappa[J]Idl \Rightarrow \frac{dI}{I} = -\kappa[J]dl$$

$$\int_{I_0}^{I} \frac{dI}{I} = -\kappa[J]\int_0^l dl \Rightarrow \ln\frac{I}{I_0} = -\kappa[J]l$$

$$\ln\frac{I}{I_0} = -\kappa[J]l \Rightarrow \log\frac{I}{I_0} = -\log e\kappa[J]l = -\varepsilon[J]l$$

- Absorbance

$$\boxed{A(\tilde{\nu}) = \log\frac{I_0}{I} = \varepsilon(\tilde{\nu})[J]l}$$

Atkins, Physical Chemistry (8th ed.) 432-433

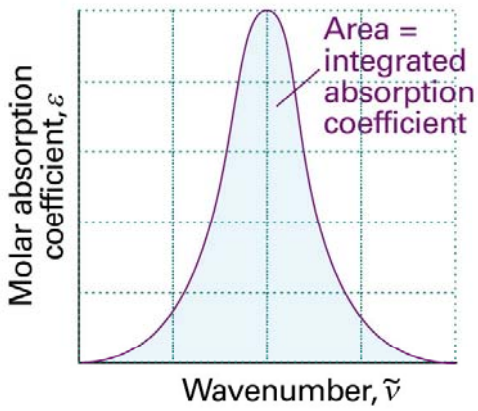


Fig. 13.4 The integrated absorption coefficient of a transition is the area under a plot of the molar absorption coefficient against the wavenumber of the incident radiation.

Colors and Complementary Colors

The colors of the visible light

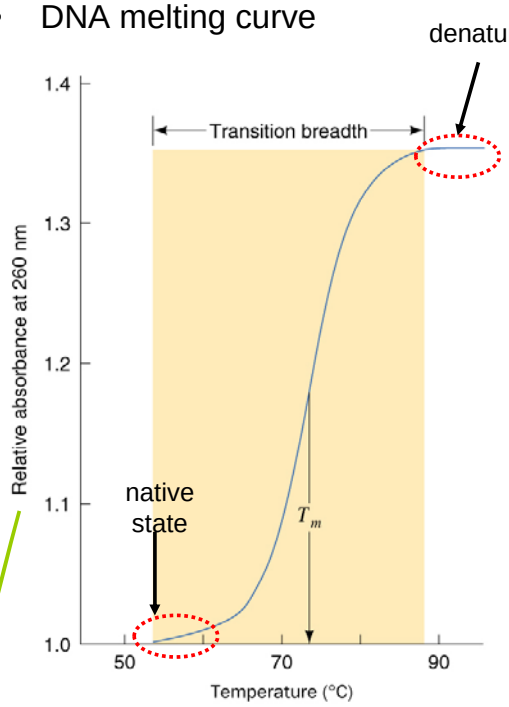
color	wavelength interval	frequency interval
red	700–635 nm	430–480 THz
orange	635–590 nm	480–510 THz
yellow	590–560 nm	510–540 THz
green	560–490 nm	540–610 THz
blue	490–450 nm	610–670 THz
violet	450–400 nm	670–750 THz

Complementary colors



Hyperchromic effect of DNA

- DNA melting curve



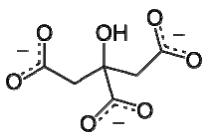
Relative absorbance: the ratio of the absorbance at the indicated temperature to that at 25°C.

- Extinction coefficients of ss- and dsDNA at 260 nm
  1. For ssDNA: 0.027 $(\mu g/ml)^{-1}cm^{-1}$
  2. For dsDNA: 0.020 $(\mu g/ml)^{-1}cm^{-1}$

- Hyperchromic effect of DNA: when DNA denatures, its UV absorbance, which is almost entirely due to its aromatic bases, increases by ~40 % at all wavelengths.

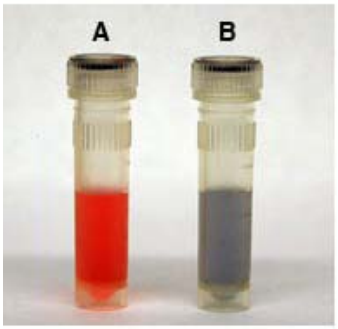
Gold nanoparticle synthesis

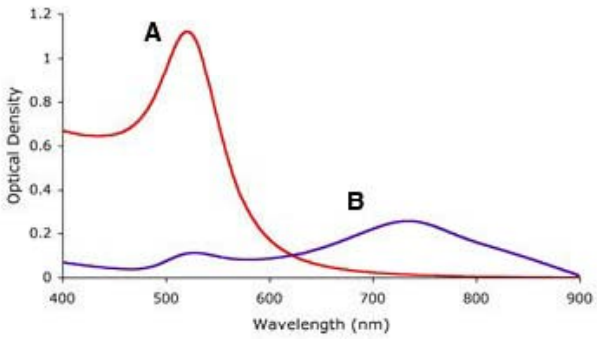
- Turkevich method
  - $HAuCl_4$ + sodium citrate
  - citrate: reducing agent and capping agent (or stabilizer)



citrate

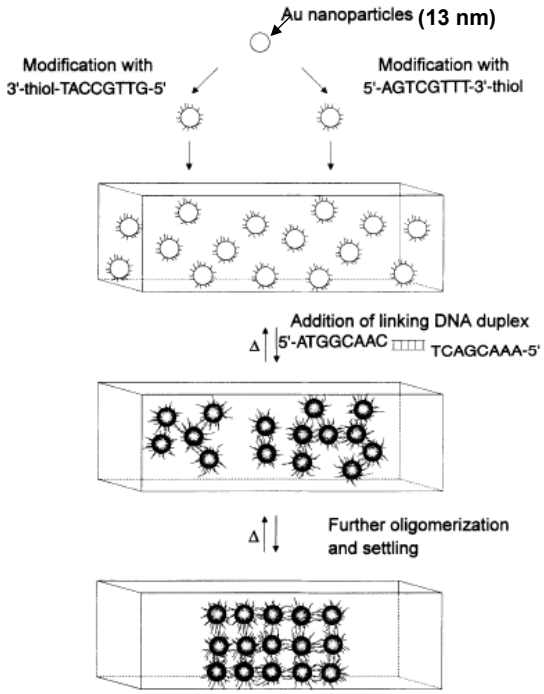
- Aggregation of gold NP as a solution of salt added

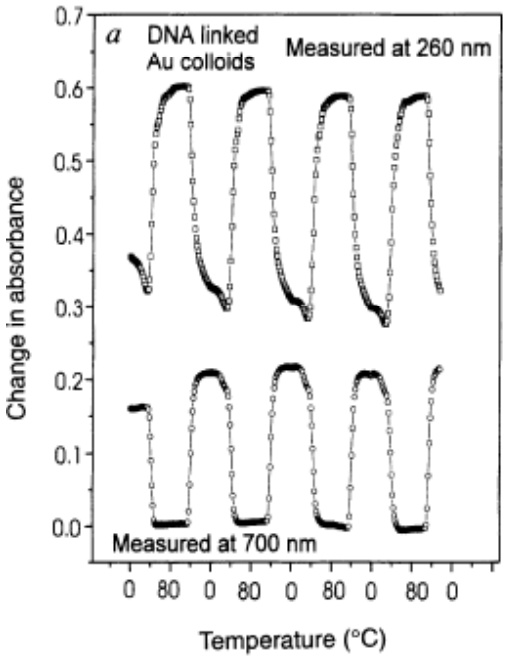




  - 금 나노입자가 떨어져 있을 때 → **522 nm** 흡수, 붉은색
  - 금 나노입자가 뭉쳐있을 때 → **700 nm** 흡수, 파랑 또는 보라색

Selective colorimetric detection of DNA

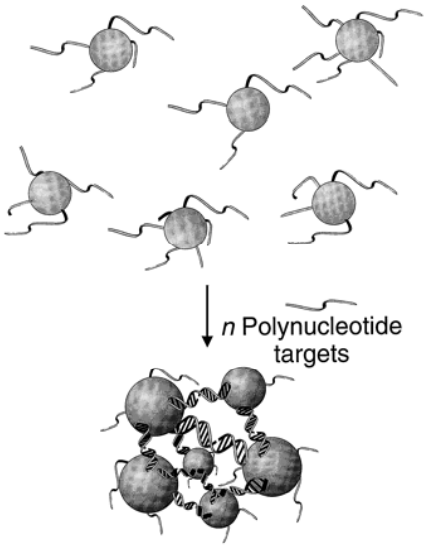




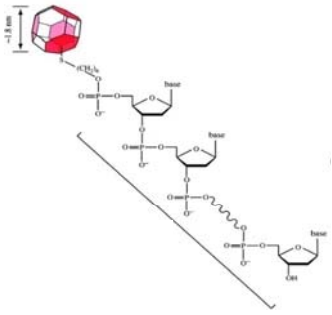
Mirkin et al., Nature **382**, 607 (1996)

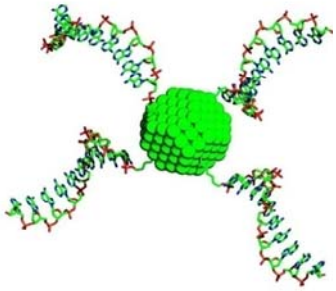
Selective Colorimetric Detection of DNA

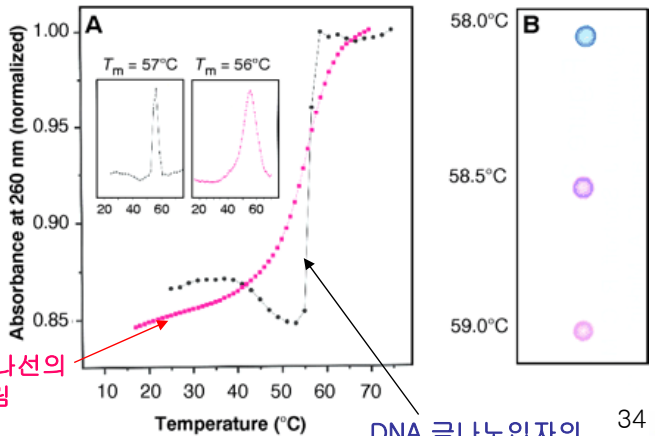
높은 온도 (58.5°C 이상)



낮은 온도 (58.5°C 이하)





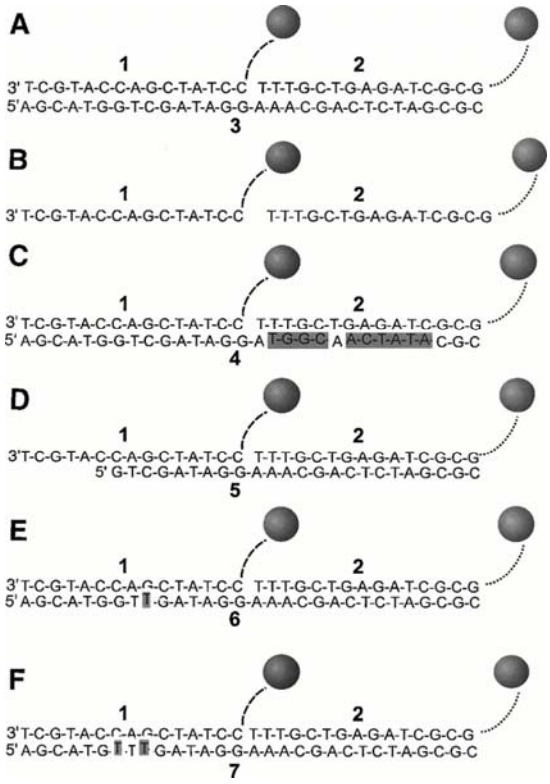


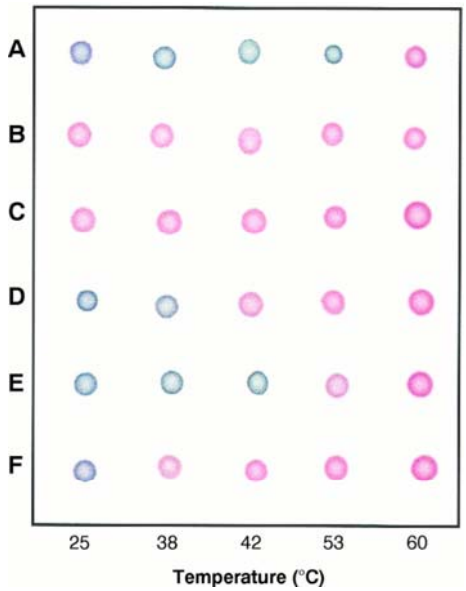
일반 DNA 이중나선의 상전이 그림

DNA 금나노입자의 상전이 그림

Mirkin et al., Science 277, 1078 (1997)

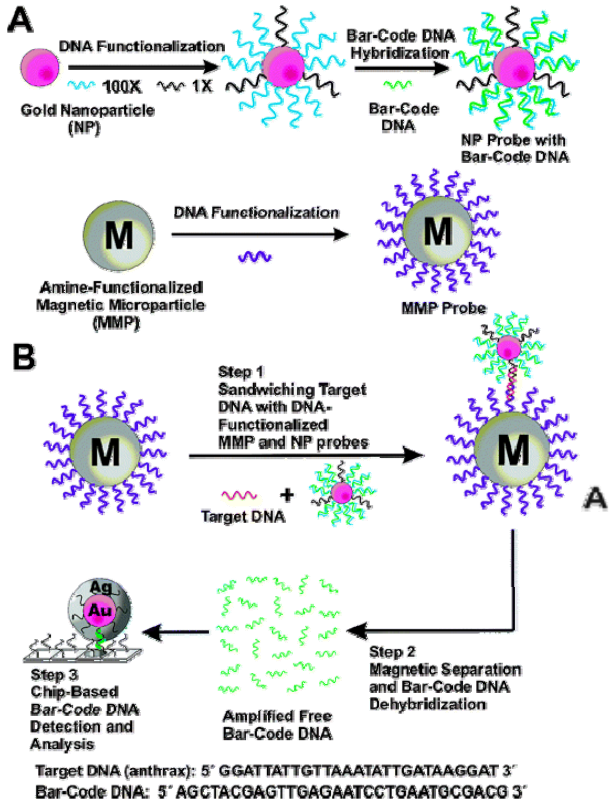
Selective Colorimetric Detection of DNA

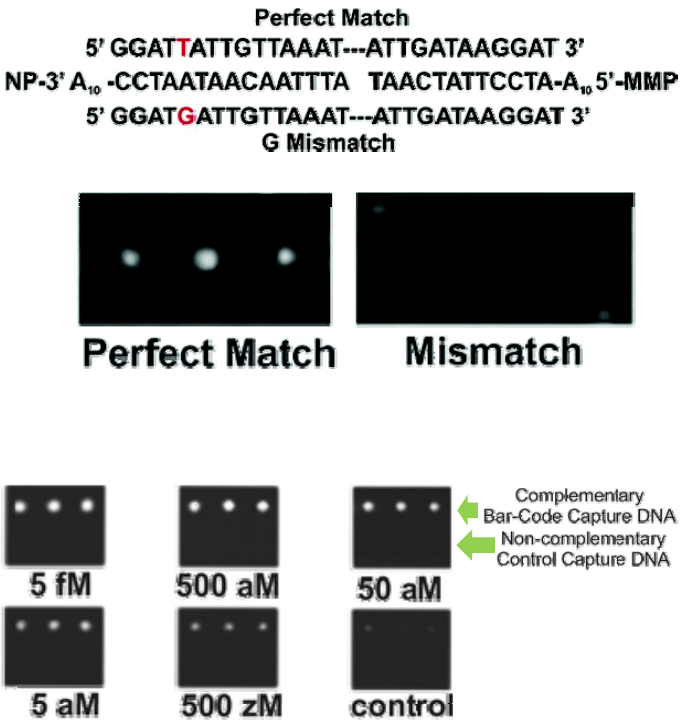




Mirkin et al., Science 277, 1078 (1997)

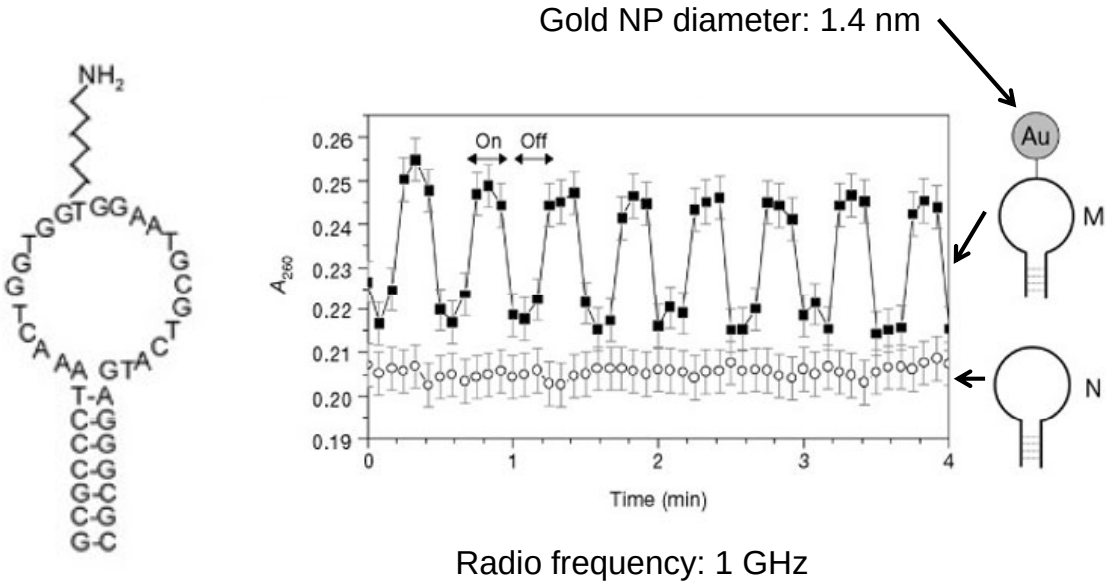
Bio-Bar-Code-Based DNA Detection





Mirkin et al., JACS **126**, 5932 (2004)

Remote Electronic Control of DNA Hybridization through Nanoscale Antenna

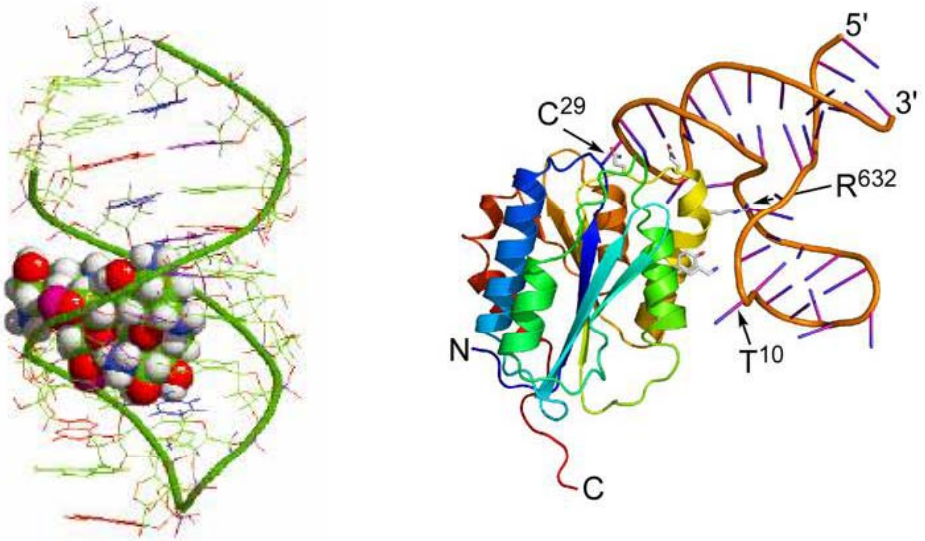


Radio frequency: 1 GHz

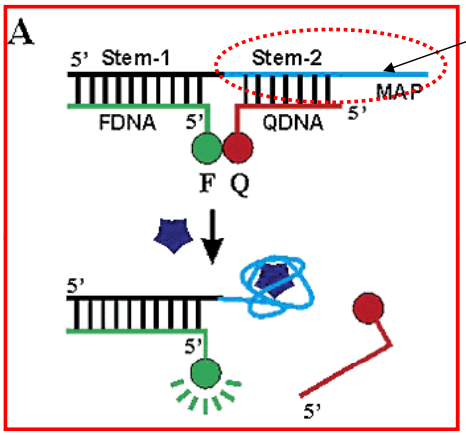
Hamad-Schifferli et al., Nature **415**, 152 (2002)

Aptamer

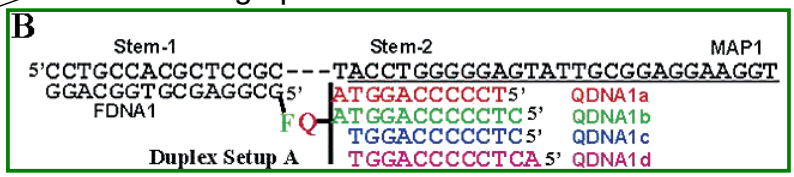
Aptamer: 핵산, 단백질, 금속이온과 같은 생체 내의 주요물질에 높은 결합력과 선택성을 가진 단일사슬 핵산 (single-stranded DNA)이나 단백질로서, 단백질로 이루어진 항체와는 달리 화학적, 물리적, 생물학적 안정성이 크다. (aptamer means “to fit” in Greek)

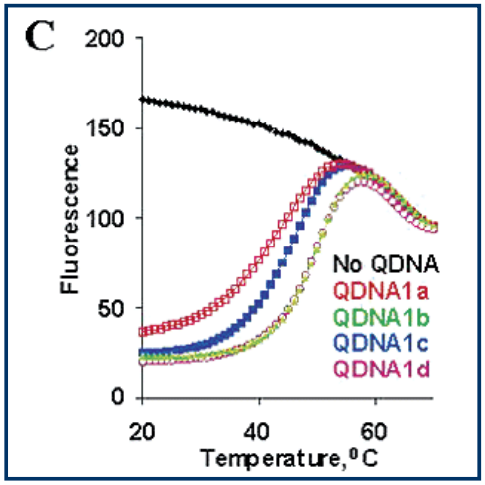


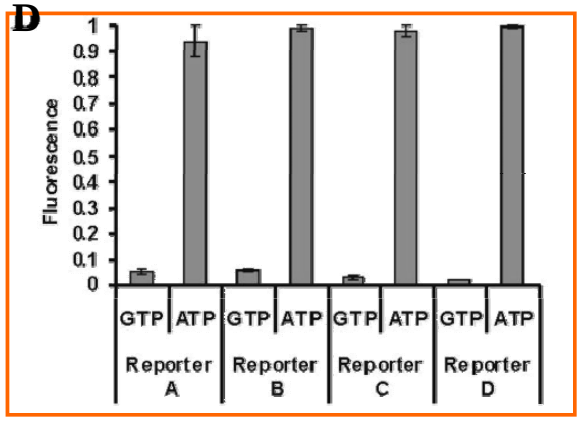
Colorimetric sensing of adenosine based on aptamers



ATP binding aptamer



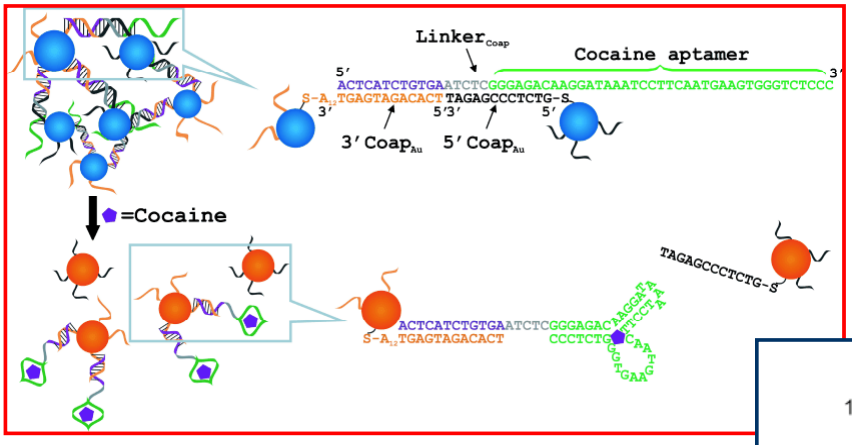


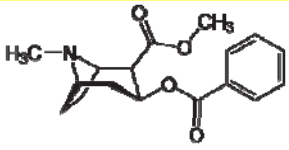


Nutiu and Li, JACS **125**, 4771 (2003)

KNU

Colorimetric sensing of cocaine based on aptamers attached to gold nanoparticles

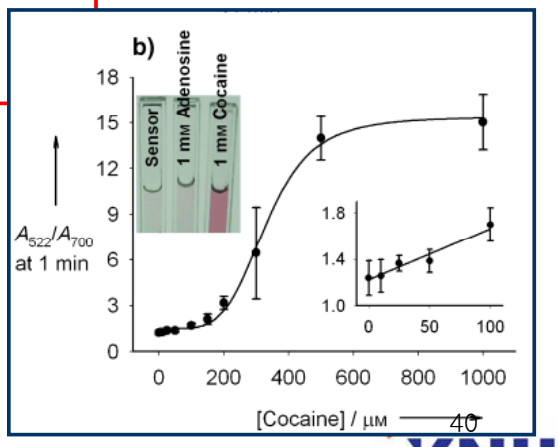




cocaine

DNA 금나노입자가 뭉쳐있을 때 → 700 nm 흡수, 파랑 또는 보라색

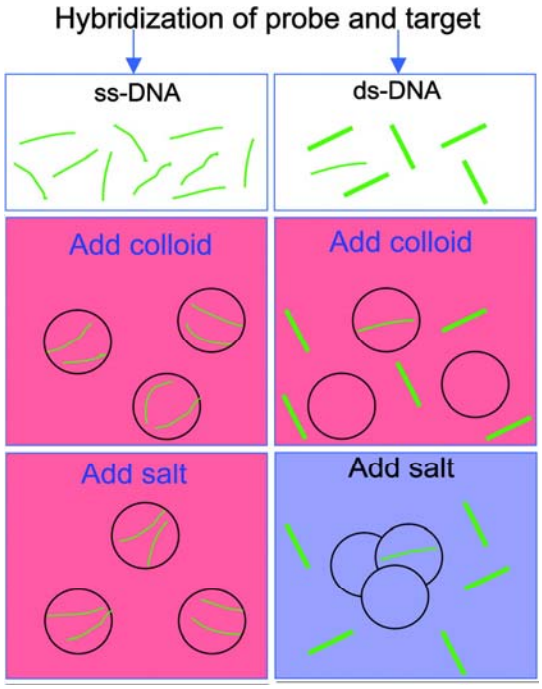
DNA 금나노입자가 떨어져 있을 때 → 522 nm 흡수, 붉은색

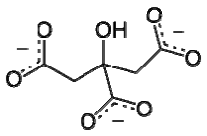


Liu and Lu, Angew. Chem. Int. Ed. **45**, 90 (2006)

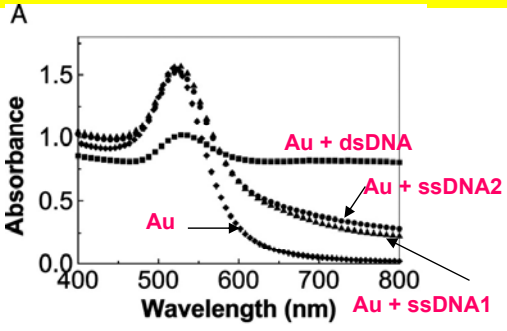
KNU

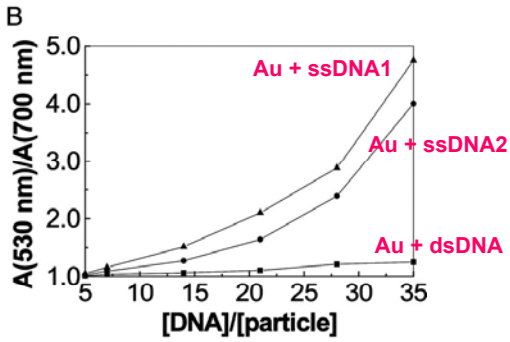
Colorimetric detection of DNA sequences based on electrostatic interactions

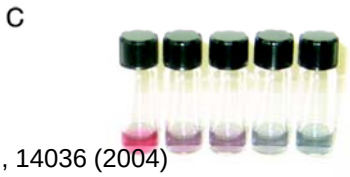




citrate

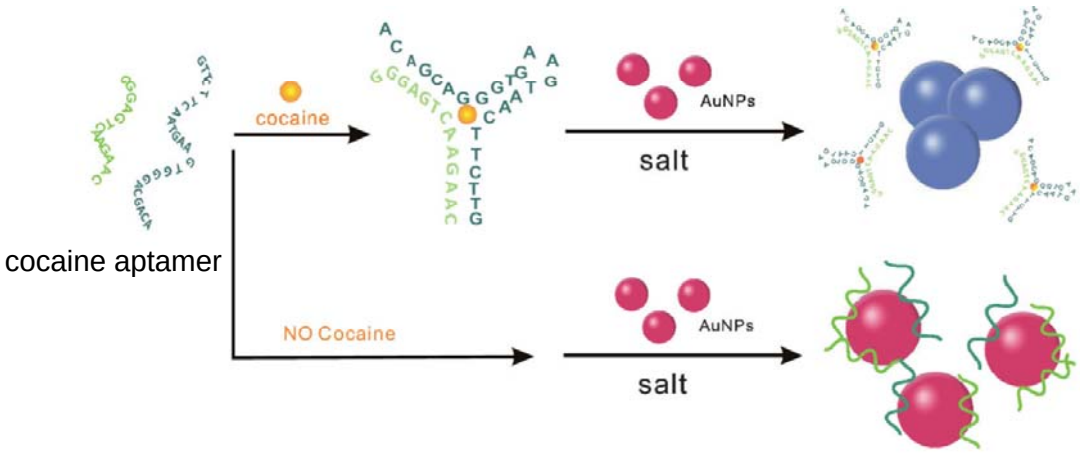






Li & Rothberg, PNAS **101**, 14036 (2004)

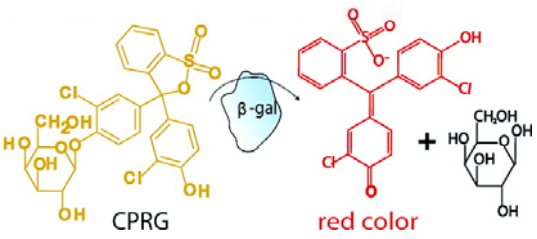
Colorimetric Sensing of Cocaine Based on Aptamers and Gold Nanoparticles



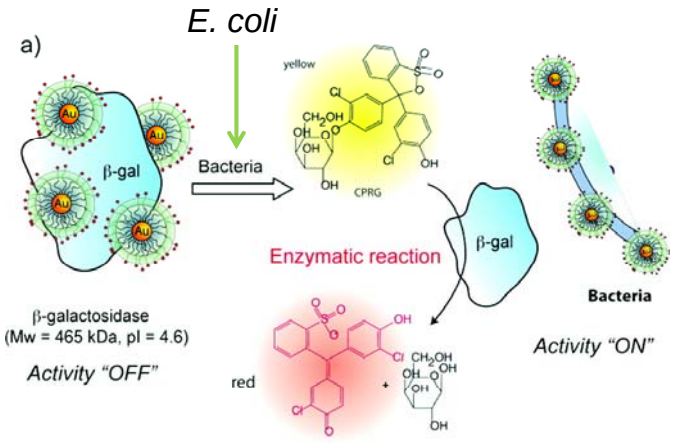


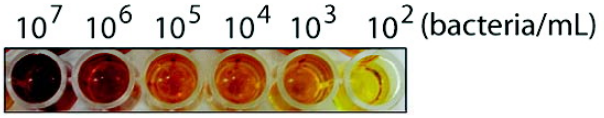
Zhang et al., Small **4**, 1196 (2008)

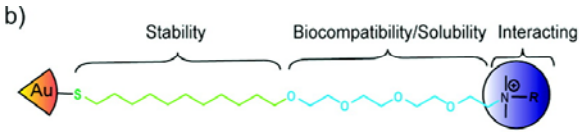
Colorimetric Sensing of Bacteria Using Enzyme-Nanoparticle Biosensor



β-galactosidase (β-gal) :효소
CPGR = β-D-galactopyranoside: 기질







Miranda et al., J. Am. Chem. Soc. **133, 9650 (2011).**